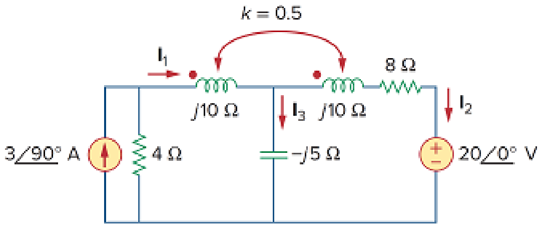
Chapter 13, Problem 20P

Problem

Determine currents **I**1, **I**2, and **I**3 in the circuit of Fig. 13.89. Find the energy stored in the coupled coils at *t* = 2 ms. Take *ω* = 1,000 rad/s.

Figure 13.89



Step-by-step solution

Step 1 of 13

Refer to Figure 13.89 in the textbook.

Apply source transformation and convert current source into voltage source, and replace $4\ \Omega$ resistance in series with voltage source.

Draw the circuit as shown in Figure 1.

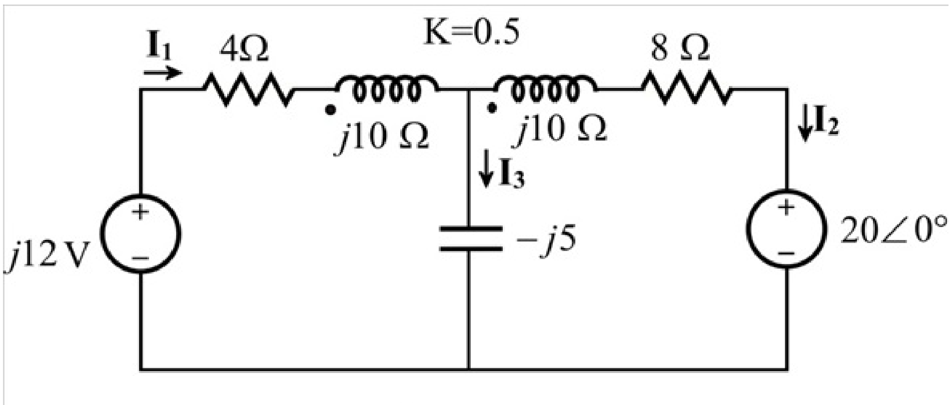


Figure 1

Step 2 of 13

Consider $K = 0.5$, $\omega = 1000$ rad/sec.

Determine the value of M.

$$K = \frac{M}{\sqrt{L_1 L_2}}$$

$$M = K\sqrt{L_1 L_2} \quad \ldots\ldots (1)$$

Determine the values of L_1.

$$j\omega L_1 = j10$$
$$\omega L_1 = 10$$
$$(1000)L_1 = 10$$
$$L_1 = 0.01$$

Step 3 of 13

Determine the value of L_2.

$$j\omega L_2 = j10$$
$$\omega L_2 = 10$$
$$(1000)L_2 = 10$$
$$L_2 = 0.01$$

Replace 0.5 for K, 0.01 for L_1, L_2 in equation 1.

$$\begin{aligned} M &= 0.5\sqrt{(0.01)(0.01)} \\ &= 0.5(0.01) \\ &= 0.005 \end{aligned}$$

Determine the value of $j\omega$.

$$\begin{aligned} j\omega M &= j(1000)(0.005) \\ &= j5 \end{aligned}$$

Step 4 of 13

Apply mesh analysis in loop 1.

$$\begin{aligned} j12 &= (4 + j10)\mathbf{I}_1 - (j5)(\mathbf{I}_1 - \mathbf{I}_2) + j5\mathbf{I}_2 \\ &= (4 + j10 - j5)\mathbf{I}_1 + j5\mathbf{I}_2 + j5\mathbf{I}_2 \\ &= (4 + j5)\mathbf{I}_1 + j10\mathbf{I}_2 \end{aligned}$$

$j12 = (4 + j5)\mathbf{I}_1 + j10\mathbf{I}_2$ ...... (2)

Apply mesh analysis in loop 2.

$$20 + (8 + j10)\mathbf{I}_2 - j5(\mathbf{I}_2 - \mathbf{I}_1) + j5\mathbf{I}_1 = 0$$
$$-20 = (8 + j10 - j5)\mathbf{I}_2 + j10\mathbf{I}_1$$

$-20 = (8 + j5)\mathbf{I}_2 + j10\mathbf{I}_1$ ...... (3)

Step 5 of 13

Solve equation 2 and equation 3.

Consider,

$$\begin{bmatrix} j12 \\ -20 \end{bmatrix} = \begin{bmatrix} 4 + j5 & +j10 \\ +j10 & 8 + j5 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

Determine the value of Δ.

$$\begin{aligned} \Delta &= (4 + j5)(8 + j5) - (j10)^2 \\ &= 32 + j20 + j40 + j^2 25 - (j^2 100) \\ &= 32 + j20 + j40 - 25 + 100 \\ &= 107 + j60 \end{aligned}$$

Step 6 of 13

Determine the value of Δ_1.

Consider the matrix as,

$$\begin{bmatrix} j12 & +j10 \\ -20 & 8+j5 \end{bmatrix}$$

$$\begin{aligned}\Delta_1 &= (j12)(8+j5)-[(-20)(j10)] \\ &= 96j + j^2 60 + j200 \\ &= -60 + j296\end{aligned}$$

Step 7 of 13

Determine the value of Δ_2.

Consider the matrix as,

$$\begin{bmatrix} 4+j5 & +j12 \\ j10 & -20 \end{bmatrix}$$

$$\begin{aligned}\Delta_2 &= [-20(4+j5)]-[j10(j12)] \\ &= -80 - j100 + 120 \\ &= 40 - j100\end{aligned}$$

Step 8 of 13

Determine the value of current $\mathbf{I}_1$.

Consider the equation for $\mathbf{I}_1$.

$\mathbf{I}_1 = \frac{\Delta_1}{\Delta}$ …… (4)

Replace $-60 - j296$ for Δ_1, and $107 + j60$ for Δ in equation 4.

$$\begin{aligned}\mathbf{I}_1 &= \frac{-60 + j296}{107 + j60} \\ &= 2.462\angle 72.18^\circ \text{ A}\end{aligned}$$

Therefore, the value of current $\mathbf{I}_1$ is $\boxed{2.462\angle 72.18^\circ}$.

Step 9 of 13

Determine the value of current $\mathbf{I}_2$.

Consider the equation for $\mathbf{I}_2$.

$\mathbf{I}_2 = \frac{\Delta_2}{\Delta}$ …… (5)

Replace $40 - j100$ for Δ_2, and $107 + j60$ for Δ in equation 5.

$$\begin{aligned}\mathbf{I}_2 &= \frac{40 - j100}{107 + j60} \\ &= 0.878\angle -97.48^\circ \text{ A}\end{aligned}$$

Therefore, the value of current $\mathbf{I}_2$ is $\boxed{0.878\angle -97.48^\circ}$.

Step 10 of 13

Determine the current $\mathbf{I}_3$.

$$\mathbf{I}_1 = \mathbf{I}_3 + \mathbf{I}_2$$

$$\mathbf{I}_3 = \mathbf{I}_1 - \mathbf{I}_2$$

Replace $2.462\angle 72.18°$ for $\mathbf{I}_1$, and $0.878\angle -97.48°$ for $\mathbf{I}_2$ in equation.

$$\begin{aligned}\mathbf{I}_3 &= 2.462\angle 72.18° - 0.878\angle -97.48° \\ &= 3.329\angle 74.89°\end{aligned}$$

Therefore, the value of the current $\mathbf{I}_3$ is $\boxed{3.329\angle 74.89°}$.

Step 11 of 13

Determine the current in time domain.

Consider $t = 2\text{ ms}$ and $\omega = 1000\text{ rad/sec}$.

$$\begin{aligned}\omega t &= (1000\text{ rad/sec})(2\text{ ms}) \\ &= 1000 \times 2 \times 10^{-3} \\ &= 2\text{ rad}\end{aligned}$$

Consider $1\text{ rad} = \frac{180°}{\pi}$.

$$\begin{aligned}2\text{ rad} &= 2\left(\frac{180°}{\pi}\right) \\ &= 114.6°\end{aligned}$$

Step 12 of 13

Consider the general equation for current in time domain.

$i(t) = A\cos(\omega t + \phi)$ ...... (6)

Replace 2.462 for A, $114.6°$ for ωt, and $72.18°$ for ϕ in equation 6.

$$\begin{aligned}i_1(t) &= 2.462\cos(114.6° + 72.18°) \\ &= -2.445\text{ A}\end{aligned}$$

Replace 0.878 for A, $114.6°$ for ωt, and $-97.48°$ for ϕ in equation 6.

$$\begin{aligned}i_2(t) &= 0.878\cos(114.6° - 97.48°) \\ &= 0.839\text{ A}\end{aligned}$$

Step 13 of 13

Determine the total energy stored in coupled coils.

Consider both current are entering.

Consider the equation for energy stored in coupled circuits.

$$W = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + M i_1 i_2 \quad \text{...... (7)}$$

Replace 0.01 for L_1, L_2, -2.455 for i_1, 0.839 for i_2, and 0.005 for M in equation 7.

$$\begin{aligned} W &= \frac{1}{2}(0.01)(-2.445)^2 + \frac{1}{2}(0.01)(0.839)^2 + (0.005)(-2.445)(0.839) \\ &= 0.0298 + \left(3.51\times10^{-3}\right) - 0.01 \\ &= 0.0232 \text{ J} \end{aligned}$$

Therefore, the energy stored in the coupled coils is $\boxed{0.0232 \text{ J}}$.